

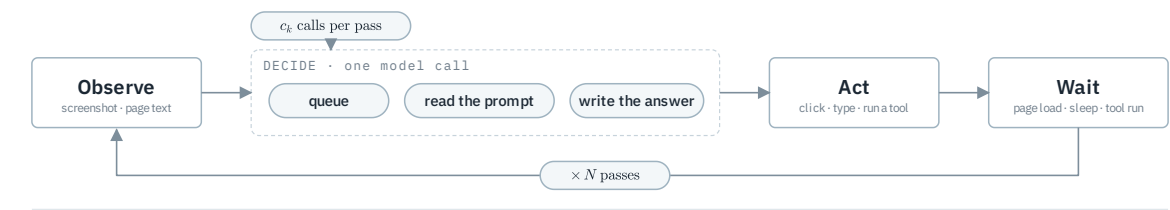
Where the time and the money go

Why LLM agents that operate software are slow and expensive, and how the two are linked.

Every number comes from a published measurement of a computer-use, web or coding agent, or from a vendor’s own price page (2024–2026); the conditions of each measurement are stated next to it, and the full references are at the end of the part.

September 2026

The loop every agent runs — Part 1 puts a number on each term

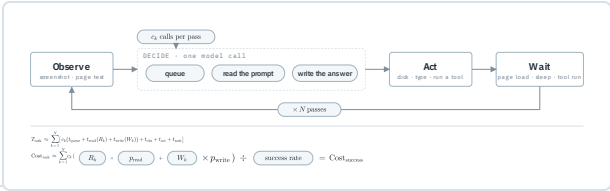


$$T_{\text{task}} \approx \sum_{k=1}^N [c_k (t_{\text{queue}} + t_{\text{read}}(R_k) + t_{\text{write}}(W_k)) + t_{\text{data}} + t_{\text{act}} + t_{\text{wait}}]$$
$$\text{Cost}_{\text{task}} \approx \sum_{k=1}^N c_k (R_k \times p_{\text{read}} + W_k \times p_{\text{write}}) \div \text{success rate} = \text{Cost}_{\text{success}}$$

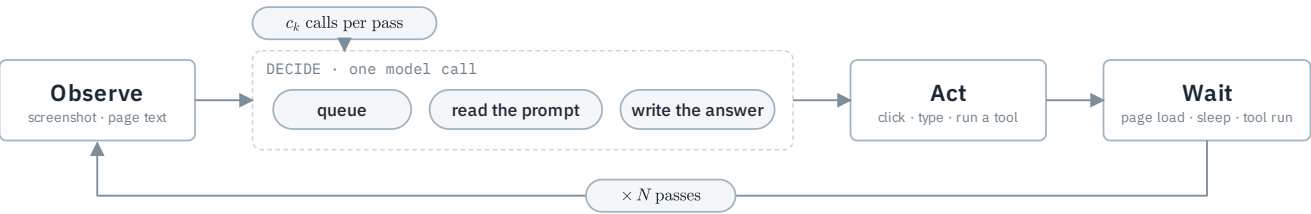
read: the prompt is processed all at once · write: the answer comes out one token at a time · N passes, c model calls per pass

The loop, with every term on it

here: the loop, its two equations, and the one term that grows fastest → to: the five sources of slowness



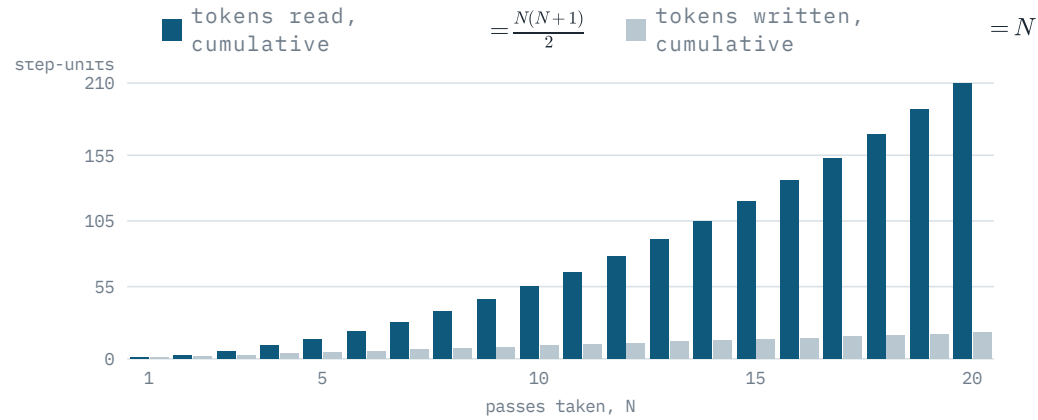
One task = one loop run N times. Each pass observes the screen, makes at least one model call, acts, and waits. A call has three segments: **queueing**; **reading** the prompt, all tokens at once; **writing** the answer, one token at a time — 81.7 tokens per second for Claude Opus 5.5 (Artificial Analysis, 2026) — at 5× the read price. **Observation, action, waiting and queueing enter only the time equation; read and write tokens enter both.**



TIME
$$T_{\text{task}} \approx \sum_{k=1}^N [c_k(t_{\text{queue}} + t_{\text{read}}(R_k) + t_{\text{write}}(W_k))] + t_{\text{obs}} + t_{\text{act}} + t_{\text{wait}}$$

MONEY
$$\text{Cost}_{\text{task}} \approx \sum_{k=1}^N c_k(R_k p_{\text{read}} + W_k p_{\text{write}}), \quad \text{Cost}_{\text{success}} = \frac{\text{Cost}_{\text{task}}}{\text{success rate}}$$

N passes · c_k model calls in pass k · R_k, W_k tokens read and written · p prices (a cached read costs 0.1×) · $t_{\text{obs}}, t_{\text{act}}, t_{\text{wait}}$ environment time. The **harness** — the program around the model — sets c_k , the observation size and the sleeps.

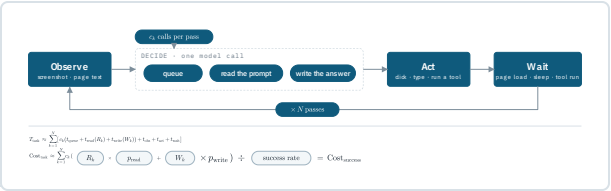


READING GROWS WITH THE SQUARE OF THE PASSES
10 passes re-read 55 units of history, 20 passes 210 (calc.)

Pass k re-sends the $k - 1$ earlier screenshots (1,000–1,800 tokens each; Anthropic, 2026a): doubling the passes quadruples the reading and only doubles the writing. Caching cuts the read price to 0.1× but not the shape. Measured: cost “grows quadratically with the number of steps”, later steps up to 3× slower (Abhyankar, Qi & Zhang, 2026).

Cumulative tokens over a task, in units of “tokens added per pass”; full history re-sent, no cache (calc.).

Slowness has five sources

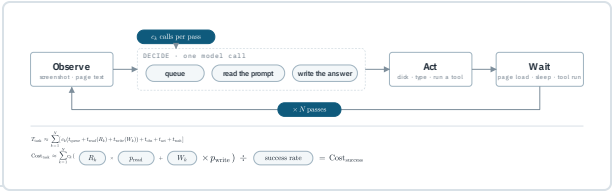


from the time equation → here: every cause of slowness, in five classes → to: the numbers behind each class

Four of the five are terms of the time equation; the fifth is the structure that adds them up. Which class dominates depends on the harness, not on the kind of agent (page 6). Cause-by-cause tables with measurement conditions: Appendix A1.

$T_{task} \approx \left \sum_{k=1..N} \right $ I · PASSES	$\left c_k \times \right $ II · CALLS PER PASS	$\left (t_{queue} + t_{read} + t_{write}) \right $ III · EACH CALL	$\left + t_{obs} + t_{act} + t_{wait} \right $ IV · ENVIRONMENT	$\left \right $ V · SERIAL
I · TOO MANY PASSES (N) 318 tool calls per task OSWorld 2.0, 108 hour-scale desktop tasks, Claude Opus 4.7, one action per step. (XLANG Lab, 2026)	II · TOO MANY CALLS PER PASS (c_k) 4–12 planning calls per judging call GTA1 harness: plan, retry, judge. Sampling candidates multiplies the calls again. (Abhyankar, Qi & Zhang, 2026)	III · EACH CALL IS SLOW queue up to 69× · read up to 3× slower by the late passes · write 91–98.6% of model time Identical requests by time of day; screenshots accumulate in the prompt; token-by-token writing. (Bian et al., 2025; Abhyankar, Qi & Zhang, 2026; Yuan et al., 2026)	IV · THE ENVIRONMENT IS SLOW 3–26 s per observation · 6.6 s per browser action · 2–3 s sleep per action Accessibility tree, browser execution, fixed sleeps; tool runs: 4% of calls take 85% of tool time. (Abhyankar, Qi & Zhang, 2026; Wong et al., 2026; Zhu et al., 2026)	V · EVERYTHING IS SERIAL concurrency within a turn: 1.15 Nothing overlaps: task time is the sum of the classes above, so fixing one term saves only its share. (Liu et al., 2026)

I–II · Too many passes, too many calls per pass



from the five classes → here: the two multipliers, passes and calls per pass → to: what happens inside a call and around it

N multiplies everything and c_k multiplies N . Passes pile up because tasks are long, each pass does one action, navigation-only passes still go through the model, and the agent idles or loops; calls pile up because the harness plans, judges and reflects, or samples several candidates per step.

I-1, I-2 · LONG TASKS, ONE ACTION PER PASS

318 tool calls per task

OSWorld 2.0, 108 long tasks, Claude Opus 4.7, one action per step. Several actions per call: 160.7 for the same model, 103 for Claude Opus 4.8.

(XLANG Lab, 2026)

I-3 · NAVIGATION-ONLY PASSES

66.7% of steps are pure navigation

Median of 151 WebVoyager tasks on live websites; three text agents driven by GPT-4o.

(Wong et al., 2026)

I-4 · IDLING AND DEAD LOOPS

66% of steps wasted · one loop: 27 min, \$8.47

GTA1 harness on 39 OSWorld tasks: failed tasks past 50 steps; one element-locating loop repeated a step 18 times, at list price without caching.

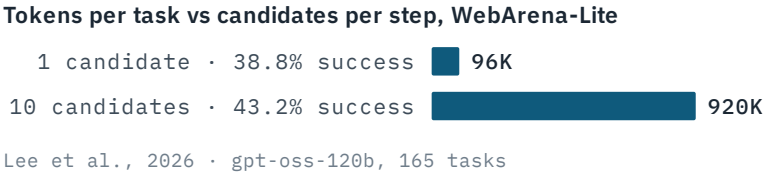
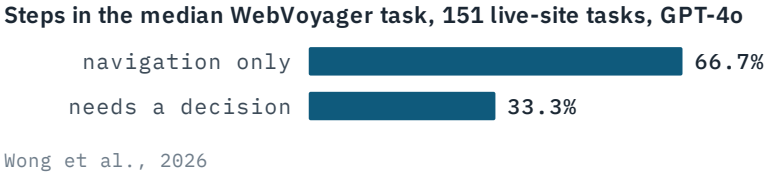
(Abhyankar, Qi & Zhang, 2026)

II · PLAN + JUDGE + REFLECT, OR SAMPLE

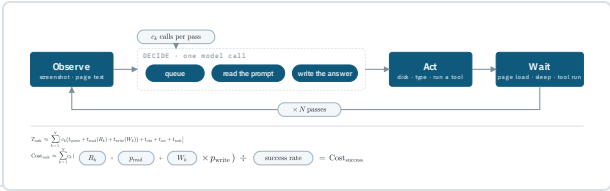
4–12 planning calls per judging call · 9.6× tokens for 4.4 points

GTA1: four parallel planners, up to three retries, one judge. Sampling 1 → 10 candidates per step: 38.8% → 43.2% success, 96K → 920K tokens (gpt-oss-120b, 165 WebArena-Lite tasks). GitHub Copilot: 6.6 model calls per user turn.

(Abhyankar, Qi & Zhang, 2026; Lee et al., 2026; Liu et al., 2026)



III–V · Inside a call, around it, and in sequence



from the multipliers → here: the per-call and per-pass terms → to: which term dominates

Inside a call: queueing, reading, writing. Around it: observation, execution, fixed sleeps, tool runs. All of it runs strictly in sequence, so task time is the sum, not the largest term.

III-1, III-2 · QUEUEING AND READING

up to 69× by time of day · up to 3× slower by the late passes · 7.14× when the cache is evicted

15 models at 5 providers; the prompt at pass k carries $k - 1$ screenshots; a richer observation multiplied the input 4.8×; GLM-4.6 on 8 H100s under load.

(Bian et al., 2025; Abhyankar, Qi & Zhang, 2026; Chang et al., 2026; Kang et al., 2026)

III-3, III-4 · WRITING AND THINKING

91–98.6% of model time · 224K vs 37K output tokens per task

Warm cache, local 27–31B models; Claude Opus 4.8 vs GPT-5.5 on the same 108 tasks. More reasoning effort lowered accuracy in 21 of 36 pairs (21,730 runs).

(Yuan et al., 2026; XLANG Lab, 2026; Kapoor et al., 2025)

IV · THE ENVIRONMENT

3–26 s per accessibility tree · 6.6 s browser vs 4.7 s model per step · 2–3 s sleep per action

OSWorld desktop apps; 151 WebVoyager live-site tasks (GPT-4o); OSWorld 2.0’s sleep $\times 318$ steps ≈ 16 min (calc.). Coding: 4% of tool calls take 85% of tool time.

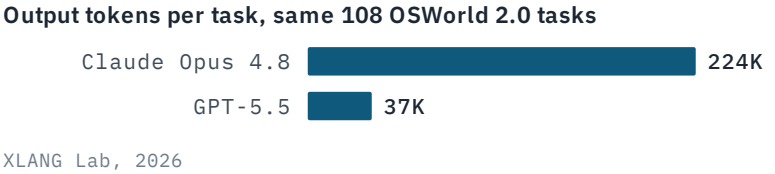
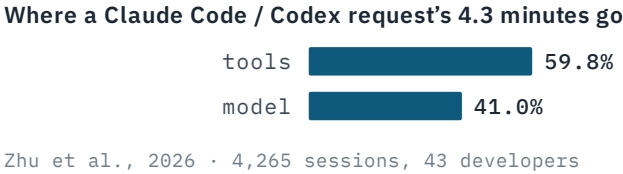
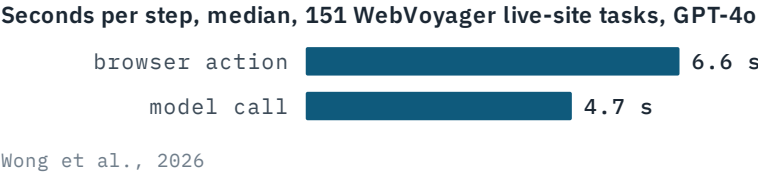
(Abhyankar, Qi & Zhang, 2026; Wong et al., 2026; xlang-ai, 2026; XLANG Lab, 2026; Zhu et al., 2026)

V · SERIAL

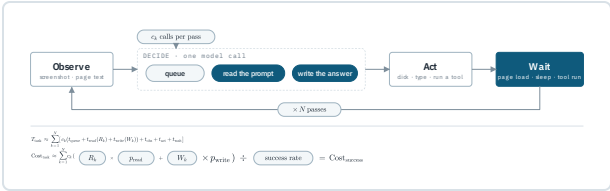
concurrency within a turn: 1.15

GitHub Copilot telemetry. Fixing one term saves only its own share. Strip the human first: 80–92% of a coding session’s elapsed time is the person thinking.

(Liu et al., 2026; Zhu et al., 2026)



Which term dominates depends on the harness, not on the kind of agent



from the five classes → here: the shares, per kind of agent → to: the six sources of cost

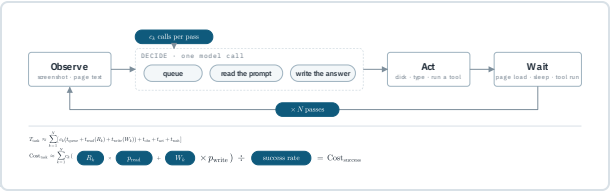
All five classes exist in every agent; the heaviest term differs, and it flips when the harness changes. The one multiplier common to all three kinds is the number of passes, {tex("N", 15)}.

	SCREENSHOT DESKTOP AGENT, MULTI-CALL HARNESS	TEXT WEB AGENT	CODING AGENT WITH A PROMPT CACHE
HEAVIEST	Reading × calls per pass × passes. 87–97% of task time in planning, judging and reflection calls; screenshots and actions under 3.5% together (GTA1, Agent S2; 39 OSWorld tasks) (Abhyankar, Qi & Zhang, 2026)	Browser execution and waiting. Per step, median: browser 6.6 s, model 4.7 s (151 WebVoyager live-site tasks, GPT-4o) (Wong et al., 2026)	Writing, or tool tails. Writing is 91–98.6% of model time (Yuan et al., 2026); Claude Code / Codex requests: tools 59.8%, model 41.0% (Zhu et al., 2026); Copilot: model 13.7%, tools 2% (Liu et al., 2026)
FLIPS WHEN	The harness makes one call per pass: over 70% of the time is then the sandbox, because the benchmark sleeps 2–3 s after every action (Chang et al., 2026; xlang-ai, 2026; XLANG Lab, 2026)	The observation grows: 4.8× input moved the model’s share from 46.9% to 61.6% (Chang et al., 2026); the Browser-Use harness spends 73% of its latency in model calls (Winston et al., 2026)	Load evicts the cache: latency up to 7.14× (Kang et al., 2026); or the tools are light and the model dominates again (Copilot)
IN ONE LINE	Slow in reading: every pass re-reads the growing screenshot history	Slow in the environment — until the observation grows, then slow in reading	Reading is cached away; slow in writing and in the tools

1 • The harness sets the dominant term — calls per pass, observation size, fixed sleeps, cache on or off. “Agents are slow because inference is slow” holds only for multi-call screenshot agents.

2 • N multiplies every term — one pass fewer saves a whole pass of time and money in all three kinds; OSWorld 2.0’s authors list “fewer environment rounds” as a goal in its own right.

Cost has six sources



from where the time goes → here: every cause of cost, in six classes → to: four ways of counting, and the machines

Money comes from two things only — tokens read and tokens written, each times a price. Writing costs 5× reading; a cached read costs 0.1×; fast mode costs 2× (Anthropic, 2026b; OpenAI, 2026). Three of the other sources multiply those two tokens; the last is the one nobody counts.

Cost ≈

$\sum_{k=1..N} c_k$
3 · NUMBER OF CALLS

$(R_k$
1 · TOKENS READ

$+ W_k \cdot 5$
2 · TOKENS WRITTEN

$) \times p$
5 · UNIT PRICE

$\div \text{success rate}$
4 · FAILURES, RETRIES

$+ \text{machine hours}$
6 · THE MACHINES

1 · TOKENS READ

quadratic in the passes

1,000–1,800 tokens per screenshot; rich observations 4.8×; sampling 96K → 920K; even at a 95.7% cache hit rate, prefix tokens are 59.5% of the bill.

(Abhyankar, Qi & Zhang, 2026; Anthropic, 2026a; Chang et al., 2026; Lee et al., 2026; Zhu et al., 2026)

2 · TOKENS WRITTEN

5× the price, billed while thinking — yet 31% of an uncached bill

224K vs 37K output tokens per task (Claude Opus 4.8 vs GPT-5.5); GTA1 on 39 OSWorld tasks: \$2.43 output only, \$7.87 all tokens.

(XLANG Lab, 2026; Abhyankar, Qi & Zhang, 2026)

3 · NUMBER OF CALLS

$N \cdot c_k$ multiplies both bills

5–13× the calls of a single-call harness per step (GTA1, calc.); sampling multiplies by the candidate count.

(Abhyankar, Qi & Zhang, 2026; Lee et al., 2026)

4 · FAILURES AND RETRIES

\$72.4 per attempt → ≈ \$351 per success

OSWorld 2.0’s best agent completes 20.6% (calc.; an accounting conversion). A dead loop bills \$8.47; one leaderboard run cost ≈ \$40,000.

(XLANG Lab, 2026; Abhyankar, Qi & Zhang, 2026; Kapoor et al., 2025)

5 · UNIT PRICE

\$4 / \$20 vs \$0.10 / \$0.50 per million tokens

Claude Opus 5.5 vs GPT-6 Luna, read / write; fast mode 2× and it drops the cache; cache write 1.25×, read 0.1×; above 272K tokens 2× (OpenAI).

(Anthropic, 2026b; OpenAI, 2026)

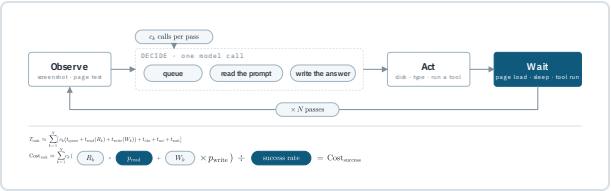
6 · THE MACHINES

0.3–4.6% of a \$7.87 bill for a one-hour CPU environment (calc.)

No benchmark reports machine dollars — every “cost per task” is an API bill; from hourly list prices, \$0.02–0.36 per hour (next page).

vendor price pages, read 2026-09-27

Counting the same task four ways, and the machines nobody counts



from the six classes → here: which convention a cost figure uses, and what the environment adds → to: how slow and expensive are linked

“What does one task cost?” has four honest answers up to 30× apart, and none of them includes the machine. The first question back is always: under which convention?

CONVENTION	THE SAME RUN, COUNTED BOTH WAYS	GAP
OUTPUT ONLY VS ALL TOKENS	GTA1 on 39 OSWorld tasks, o3 list price, no cache: \$2.43 counting output only; \$7.87 counting all tokens (Abhyankar, Qi & Zhang, 2026)	3.2×
PER ATTEMPT VS PER SUCCESS	OSWorld 2.0, Claude Opus 4.8: \$72.4 per attempt at 20.6% completion → ≈ \$351 (calc.; not the cost of retrying to success) (XLANG Lab, 2026)	4.9×
UNCACHED VS CACHED	A cached read costs 0.1×; but at a 95.7% hit rate, prefix tokens were still 59.5% of the bill (4,265 Claude Code / Codex sessions) (Anthropic, 2026b; Zhu et al., 2026)	10× read price only
STANDARD VS FAST MODE	Claude Opus 5.5 \$4 / \$20 → \$8 / \$40; OpenAI 2× on every listed model; up to 2.5× faster writing (vendor-stated), reading unchanged (Anthropic, 2026b; OpenAI, 2026)	2×

The three that apply to a whole bill compound to about 31× (3.2 × 4.9 × 2, calc.). Screenshot-agent measurements do not report their cache state.

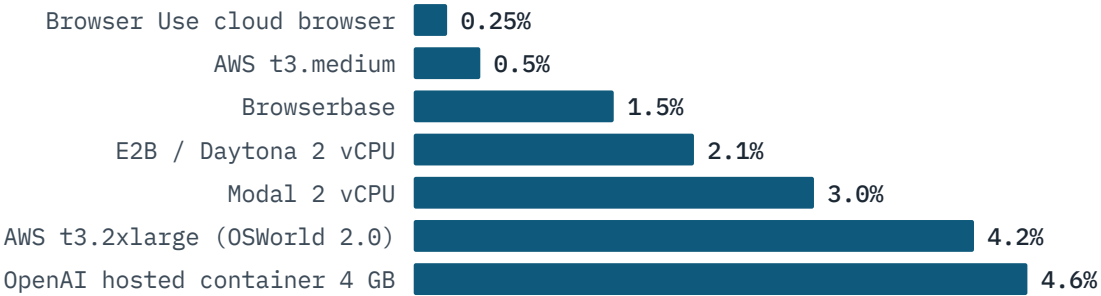
6 · THE MACHINES, FROM LIST PRICES

a one-hour CPU environment: \$0.02–0.36 → 0.3–4.6% of a \$7.87 bill, 0.03–0.5% of \$72.4 (calc.)

Benchmarks name their instances but never their cost; products (Meta Muse, ChatGPT agent) run a dedicated cloud machine per user and publish no price. It grows large with plan fees, residential-proxy traffic (\$5–12 per GB), wall-clock billing while the agent waits (the median AgentSysBench session is active 20% of its lifetime), and GPU sandboxes (an H100 at \$3.95 per hour is half of a \$7.87 bill).

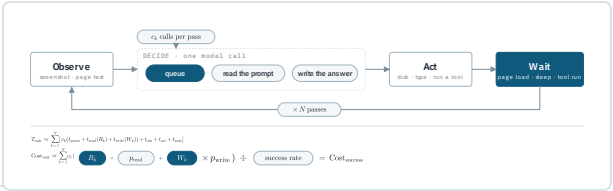
vendor price pages, read 2026-09-27; Chang et al., 2026; Meta, 2026

A one-hour CPU environment as a share of a \$7.87 API bill (calc.)



vendor price pages, read 2026-09-27 · full price table in Appendix A3

How slow and expensive are linked: three kinds of relationship



from the two bills → here: what each cause does to time and to money → to: the levers

The two equations share the read and write tokens; environment time and queueing appear only in the time equation. So every cause relates to time and money in exactly one of three ways.

TIME

$$\sum_{k=1..N} c_k \times (t_{\text{queue}} + t_{\text{read}} + t_{\text{write}}) + t_{\text{obs}} + t_{\text{act}} + t_{\text{wait}}$$

MONEY

$$\sum_{k=1..N} c_k \times (R_k \times p_{\text{read}} + W_k \times p_{\text{write}}) \div \text{success rate}$$

in both equations — same source

time equation only — slow but not expensive

the price — money for time or accuracy

SAME SOURCE — IN BOTH EQUATIONS

fix it, and time and money fall together

Passes N (time ≈ linear, read tokens quadratic), **calls per pass** c_k , **reading, writing** (the slowest and dearest token), **cache hits** (fragile: editing the history or switching to fast mode invalidates them), **failures and idle loops** (which also turn “per attempt” into “per success”).

(Abhyankar, Qi & Zhang, 2026; Anthropic, 2026b)

SLOW BUT NOT EXPENSIVE — TIME EQUATION ONLY

fix it, and only time falls

Queueing, page loads and fixed sleeps, tool tails. None produces a token; the machine behind them is billed by the hour, usually under 5% of the API bill. Escaping the queue means a fast or priority tier — at 2× the price.

MONEY FOR TIME OR FOR ACCURACY — OPPOSITE SIGNS

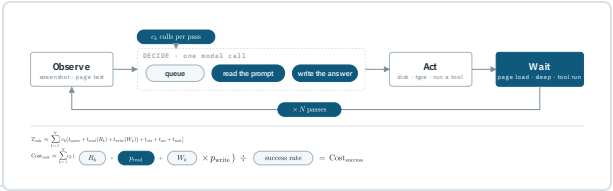
buying speed buys only the writing segment

Fast mode: up to 2.5× faster writing (vendor-stated), reading unchanged, 2× the price. **Bigger models, more thinking:** cost versus accuracy is convex — each extra point costs more than the last (marginal tokens per point 101K → 575K; 6×, 9×, 9.6× the tokens for a few points), and more thinking is not always more accurate.

(Anthropic, 2026b; OpenAI, 2026; XLANG Lab, 2026; Kapoor et al., 2025; Lee et al., 2026)

In one line: a cause that lives in tokens — fix it and time and money fall together; a cause that lives in waiting — fix it and only time falls; paying for speed buys only the writing segment.

The levers, and what is still unmeasured



from the three relationships → here: what fixing each lever buys, and the holes in the evidence → to: Part 2, what the literature has done with each lever

Seven levers follow from the equations; each saves a known term and carries a known risk. Part 2 asks how far the literature has pushed each one. On the right: what no measurement has settled — gaps in the evidence, not claims of a research gap.

LEVER	TIME	MONEY	THE CATCH
Fewer passes — several actions per call, skip navigation-only steps	yes	yes, the quadratic term	which steps need no thinking?
Read less — trim history, shrink screenshots, cache the prefix	yes	yes	forgets; trimming invalidates the cache
Write less — less thinking, smaller model	yes	yes	accuracy may fall
Fewer calls per pass — drop judging and reflection	yes	yes	fewer errors caught
Overlap the waiting — pre-load, parallel tools, page-ready events	yes	no	only where the environment dominates
Buy a fast or priority tier	writing only	no — 2× more	reading unchanged; cache dropped
Stop retrying failures — early stop, detect dead loops	yes	yes	kills some attempts that would succeed

NOT YET MEASURED

- **Read / write time split and cache hit rate for a frontier API model** — the only split is on local 27–31B models (Yuan et al., 2026); screenshot-agent measurements do not report cache state.
- **How much slower than a person** — no benchmark times agent and human on the same tasks; the closest, AXIS’s small user study, has a UI agent 1.69× slower than manual work on easy tasks (Lu et al., 2025).
- **Whether the benchmark’s fixed sleep sits inside “action” time** in OSWorld-Human — if so, “environment under 3.5%” understates the waiting.
- **How much slowness costs accuracy** — OSWorld 2.0 records the stale-screen failure mode without its share (XLANG Lab, 2026); the rebound from a smaller model has no direct evidence.

References · Part 1 (1 of 2)

author-year tags on the pages refer to these entries · published version where one exists, otherwise the arXiv number and the authors’ institutions

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The diagram illustrates the architecture of an agentic inference system. It shows a flow from 'Observe' to 'Act' to 'Wait'. The 'Observe' step involves 'observe prompt, single tool'. The 'Act' step involves 'act, type, run a tool'. The 'Wait' step involves 'wait, load, sleep, tool out'. A 'queue' step is shown between 'Observe' and 'Act', with 'read the prompt' and 'write the answer' steps. A 'DECIDE - one model call' step is shown above the queue. A 'costs per pass' box is shown above the 'DECIDE' step. A 'x N passes' box is shown below the 'Wait' step. Below the diagram, the cost formula is given:
$$C_{total} = \sum_{i=1}^N (C_{prompt} + C_{tool}(R_i) + C_{tool}(R_i) + C_{wait} + C_{sleep} + C_{tool out})$$

$$Cost_{total} = \sum_{i=1}^N (R_i \cdot C_{prompt} + C_{tool} \cdot W_i \cdot N_{Passes}) + C_{sleep rate} \cdot Cost_{sleep}$$

References · Part 1 (2 of 2)

continued

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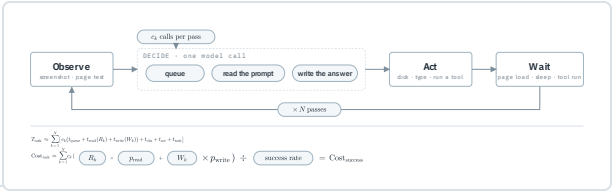
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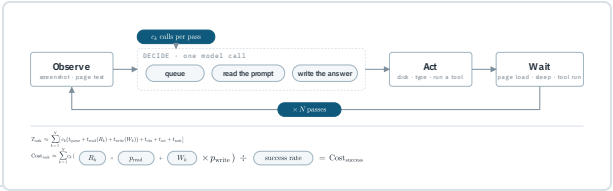
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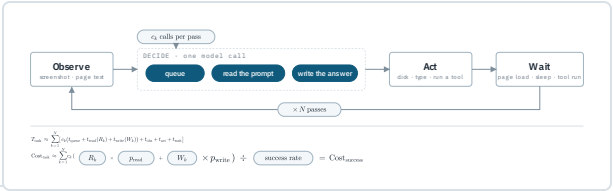
A1 · Slowness, every cause — I. steps and II. calls per step



the complete table behind pages 3–4 · three kinds of agent: screenshot desktop agent, text web agent, coding agent with a prompt cache

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE	DOMINANT IN
I-1	The task is long	Hour-scale tasks take hundreds of passes	OSWorld 2.0, 108 long tasks: Claude Opus 4.7 with maximum thinking, one action per step, 500-step cap: 318 tool calls per task on average, i.e. 318 steps; with several actions per call, Claude Opus 4.8 takes 103 steps (481.8 calls) and GPT-5.5 95.2 steps (149.8 calls)	(XLANG Lab, 2026)	screenshot, text
I-2	One action per pass	Every click is a whole pass: one observation, one model call, one wait	OSWorld 2.0: with one action per step Claude Opus 4.7 needs 318 steps; with batched actions the same model needs 160.7 and Claude Opus 4.8 needs 103	(XLANG Lab, 2026)	screenshot, text
I-3	Navigation-only passes still call the model	Paging, scrolling and opening the next page need no thinking but cost a call each	Skim’s profile of three text web agents (Browser-Use, AgentOccam, WebVoyager; GPT-4o) on 151 WebVoyager live-site tasks: 66.7% of the steps in the median task are pure navigation	(Wong et al., 2026)	text
I-4	Idling and dead loops	Element not found, repeated retries; steps rise without progress	OSWorld-Human’s records of the GTA1 harness (o3 plans and judges, GTA1-7B locates elements) on 39 OSWorld desktop tasks: in failed tasks that exceeded 50 steps, 66% of steps were wasted repeats; one element-locating loop repeated the same step 18 times — 27 minutes and \$8.47 at list price without caching	(Abhyankar, Qi & Zhang, 2026)	screenshot
I-5	Acting on a stale screen	The screenshot is taken before the page changes; the agent clicks on the old layout and must repair	OSWorld 2.0 records “stale interface state” as a failure mode; no share reported	(XLANG Lab, 2026)	screenshot
II-1	Multi-call harness	One step = plan + judge + reflect, several calls in series	OSWorld-Human, two harnesses: GTA1 runs 4 parallel planners per step, retries up to 3 rounds, then one judging call picks — 4–12 planning calls per judging call; Agent S2 adds a reflection call per step, so planning is 53% and reflection 34% of task time	(Abhyankar, Qi & Zhang, 2026)	screenshot
II-2	Sampling for accuracy	5–20 candidates per step, then pick one	Same model and harness (gpt-oss-120b, ReAct) on 165 WebArena-Lite tasks: 1 → 10 candidates per step raises success from 38.8% to 43.2% and tokens per task from 96K to 920K — 9.6× the tokens for 4.4 points	(Lee et al., 2026)	text
II-3	Coding agents call repeatedly within a turn	The model is called back and forth inside one user turn	GitHub Copilot coding agent production telemetry (first week of June 2026, 13.5 million sessions): 6.6 model calls per user turn on average	(Liu et al., 2026)	coding

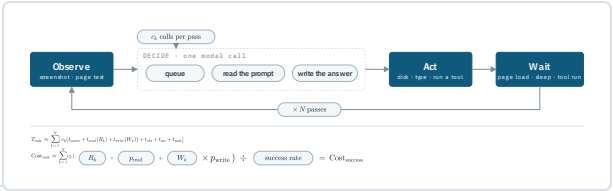
A1 · Slowness, every cause — III. each call is slow



the complete table behind page 5

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE	DOMINANT IN
III-1	Queueing	The request waits at the provider; unbilled, but counted as model time	Client-side measurement of 15 models across 5 providers: requests of the same length differ in latency by up to 69× depending on when they are sent	(Bian et al., 2025)	all
III-2	Reading (the prompt is re-processed every call)	Every call re-reads the whole history; more screenshots, slower	OSWorld-Human, multi-call harnesses: later steps up to 3× slower because the prompt at step k holds the k–1 earlier screenshots — “dominated by prefill”. One screenshot is 1,000–1,800 tokens. AgentSysBench’s WebArena agent (Kimi-K2.6): observation switched from a single format to accessibility tree + HTML + screenshot → input 4.8×, model share of time 46.9% → 61.6%. Under load the cache is evicted and re-read: SWE-Agent / OpenHands running GLM-4.6 on 8 H100s for SWE-bench Lite, request latency up to 7.14× as concurrency rises	(Abhyankar, Qi & Zhang, 2026; Anthropic, 2026a; Chang et al., 2026; Kang et al., 2026)	screenshot; text with rich observations; coding under load
III-3	Writing (one token at a time)	Token-by-token generation, strictly serial	Locally served ReAct agents (Qwen3.6-27B / Gemma4-31B, vLLM, 2 H100s) on five benchmarks: with the cache warm, generation is 91–98.6% of model time. Windows desktop agent UFO2 (GPT-4o / o1 API): about 10 s per model call, the largest item per step in every configuration	(Yuan et al., 2026; Zhang et al., 2025)	coding; single-call screenshot
III-4	Thinking modes and larger models	Thinking writes more tokens; larger models are slower per token	OSWorld 2.0, same 108 tasks: Claude Opus 4.8 writes 224K output tokens per task, GPT-5.5 37K. Holistic Agent Leaderboard, 9 benchmarks, 21,730 runs: raising reasoning effort lowered accuracy in 21 of 36 pairs	(XLANG Lab, 2026; Kapoor et al., 2025)	all

A1 · Slowness, every cause — IV. environment, V. serial



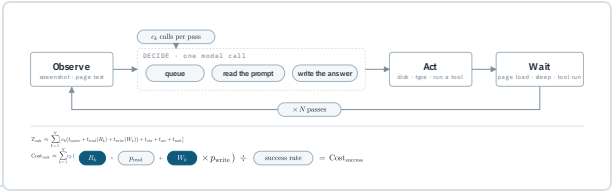
the complete table behind page 5 (environment and serial structure)

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE	DOMINANT IN
IV-1	Producing the observation	Extracting the accessibility tree or detecting elements takes time	OSWorld desktop applications: 3–26 s to generate one accessibility tree; the screenshot itself is under 2% of task time. UFO2: OmniParser element detection adds about 1 s per step	(Abhyankar, Qi & Zhang, 2026; Zhang et al., 2025)	screenshot
IV-2	Browser execution and page load	After a click the agent waits for the page	151 WebVoyager live-site tasks: per step, median browser action 6.6 s vs model call 4.7 s. WebArena with GenericAgent + Claude 3.5 Sonnet: 7.6 s of a 12.2 s step in the browser (this figure was not re-verified)	(Wong et al., 2026; Le Sellier De Chezelles et al., 2025)	text
IV-3	Fixed sleeps	The agent cannot tell when the page is ready, so it sleeps a fixed number of seconds	OSWorld’s environment code sleeps 2 s after every action; OSWorld 2.0 prescribes 3 s, so 318 steps are about 16 minutes of pure waiting (calc.). AgentSysBench: a single-call GUI agent (ReAct, Kimi-K2.6 API) spends over 70% of its OSWorld execution time in the desktop sandbox	(xlang-ai, 2026; XLANG Lab, 2026; Chang et al., 2026)	screenshot (caused by the benchmark harness)
IV-4	Tool tails	Tests and builds take minutes	TraceLab, 4,265 Claude Code / Codex sessions from 43 developers (September 2025 – June 2026): tool calls over 1 minute are 4% of calls but 85% of tool time; a request averages 4.3 minutes — tools 2.5 (59.8%), model 1.7 (41.0%)	(Zhu et al., 2026)	coding
V	Serial structure	The four stages never overlap, so task time is the sum of every term, not the largest one; fixing one term saves only its share	Concurrency within a turn is 1.15 in Copilot telemetry. Separately, 80–92% of a coding session’s elapsed time is the human thinking between turns — strip it before reading any total	(Liu et al., 2026; Zhu et al., 2026)	all



[← back](#)

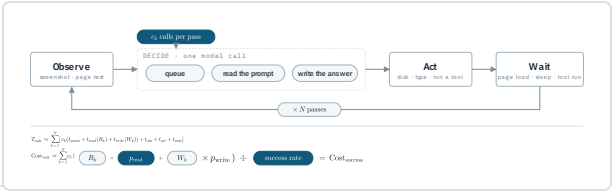
A3 · Cost, every cause — 1. tokens read and 2. tokens written



the complete table behind page 7 · prices are list prices from the vendors’ own pages, September 2026

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE
1-1	History re-sent	Call k reads everything from the $k - 1$ earlier steps; total reading $\approx N^2$	“Cost grows quadratically with the number of steps” — OSWorld-Human, real runs	(Abhyankar, Qi & Zhang, 2026)
1-2	Large screenshots	1,000–1,800 tokens per image; 100 images fill a 200K context	Anthropic’s engineering guidance for computer and browser use	(Anthropic, 2026a)
1-3	Rich observation formats	Accessibility tree + HTML + screenshot together: 4.8× the input	AgentSysBench, WebArena agent, Kimi-K2.6	(Chang et al., 2026)
1-4	Sampling	Several candidates per step, mostly read tokens	gpt-oss-120b + ReAct on 165 WebArena-Lite tasks: 1 → 10 candidates per step, 96K → 920K tokens per task	(Lee et al., 2026)
1-5	The cache is not a cure	However high the hit rate, a long history keeps reading dominant; a miss re-reads everything	TraceLab, 4,265 Claude Code / Codex sessions: prefix-cache hit rate 95.7%, prefix tokens still 59.5% of list-price cost; on a miss the re-read is 3.8× the genuinely new content	(Zhu et al., 2026)
2-1	Thinking modes	Thinking tokens are billed as output; when they cannot be switched off they are a fixed overhead	OSWorld 2.0, same tasks: Claude Opus 4.8 224K output tokens per task, GPT-5.5 37K; Claude Opus 5.5: “thinking cannot be disabled and is billed as output”	(XLANG Lab, 2026; Anthropic, 2026b)
2-2	But writing is not always the larger bill	In an uncached screenshot agent, output was 31% of the total	OSWorld-Human, GTA1: the paper’s \$2.43 counts output only; Table 3’s input and output give \$7.87	(Abhyankar, Qi & Zhang, 2026)

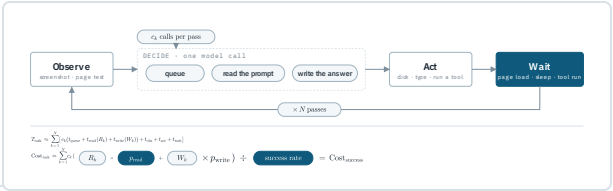
A3 · Cost, every cause — 3. calls, 4. failures, 5. unit price



the complete table behind page 7 (calls, failures, unit price)

ID	CAUSE	MECHANISM	EVIDENCE, WITH THE CONDITIONS OF THE MEASUREMENT	SOURCE
3	Number of calls	$N \cdot c_k$ scales both token bills	GTA1: 4 parallel planners per step, up to 3 retry rounds, one judging call → 4–12 planning calls per judging call, i.e. 5–13× the calls of a single-call harness (calc.). Agentic test-time scaling: 5–20 candidates per step multiply the calls by the candidate count. The same extra calls cost time and money at once	(Abhyankar, Qi & Zhang, 2026; Lee et al., 2026)
4-1	Failed attempts are billed	Cost per success = cost per attempt ÷ success rate	OSWorld 2.0: the best agent (Claude Opus 4.8) costs about \$72.4 per attempt at 20.6% completion → about \$351 per success (calc.). An accounting conversion, not the real price of retrying one task until it succeeds	(XLANG Lab, 2026)
4-2	Idle steps are billed in full	Every pass of a dead loop is a full read and write	OSWorld-Human, GTA1 on 39 OSWorld tasks: one element-locating loop repeated the same step 18 times — 27 minutes, \$8.47 at list price without caching; in failed tasks over 50 steps, 66% of steps were such repeats	(Abhyankar, Qi & Zhang, 2026)
4-3	Evaluation itself is too expensive to repeat	Single runs, no repeats	Holistic Agent Leaderboard: 9 benchmarks, 21,730 runs, one run per configuration, about \$40,000 in total; Claude Opus 4.1 not run on Online-Mind2Web because the estimate was \$20,000	(Kapoor et al., 2025)
5-1	Model tier	Claude Opus 5.5 \$4 / \$20 vs GPT-6 Luna \$0.10 / \$0.50 per million tokens read / written	Vendor price pages, 27 September 2026	(Anthropic, 2026b; OpenAI, 2026)
5-2	Fast mode	2× on both vendors; switching speed tiers invalidates the cache	Anthropic: Claude Opus 5.5 \$8 / \$40, up to 2.5× output speed (vendor-stated), first-token wait unchanged. OpenAI: 2× on all listed models; “up to 2.5×” stated only for GPT-5.6 Sol	(Anthropic, 2026b; OpenAI, 2026)
5-3	Cache write and read	Writing to the cache 1.25×; reading from it 0.1×	Vendor price pages	(Anthropic, 2026b; OpenAI, 2026)
5-4	Long-context surcharge	OpenAI: 2× the read price above 272K tokens	Vendor price page	(OpenAI, 2026)

A3 · Cost — 6. the machines, and the four conventions



the complete tables behind page 8 · vendor price pages read 2026-09-27 · per-task amounts and shares are our own arithmetic (calc.)

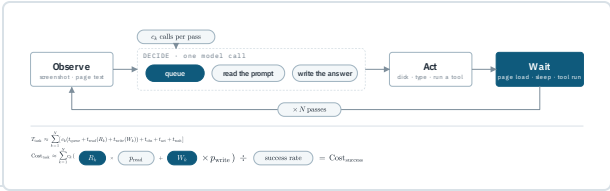
ENVIRONMENT	PRICE (VENDOR PAGE)	1-HOUR TASK	15-MIN TASK	SHARE OF A \$7.87 API BILL	SHARE OF \$72.4
Browser Use cloud browser	\$0.02 per browser-hour, billed by the minute, 1-minute minimum; traffic extra (residential proxy \$5/GB, direct \$0.20/GB); sessions capped at 240 minutes	\$0.020	\$0.005	0.25%	0.03%
AWS t3.medium (2 vCPU / 4 GiB, Linux, us-east-1)	\$0.0418 per hour on the T3 product page; another AWS page says \$0.0416 (0.5% apart)	\$0.042	\$0.010	0.5%	0.06%
Browserbase cloud browser	\$20 per month for 100 hours, then \$0.12 per hour; \$99 per month for 500 hours, then \$0.10; billed by the minute	\$0.12	\$0.03	1.5%	0.17%
E2B / Daytona sandbox (2 vCPU / 4 GiB)	E2B \$0.000014 per vCPU-second + \$0.0000045 per GiB-second; Daytona \$0.0504 per vCPU-hour + \$0.0162 per GiB-hour — both \$0.1656 per hour	\$0.166	\$0.041	2.1%	0.23%
Modal sandbox (1 physical core = 2 vCPU, 4 GiB)	\$0.00003942 per core-second + \$0.00000667 per GiB-second → \$0.238 per hour; billed on the larger of requested and used	\$0.238	\$0.060	3.0%	0.33%
AWS t3.2xlarge (8 vCPU / 32 GiB, OSWorld 2.0's default instance)	\$0.3341 per hour	\$0.334	\$0.084	4.2%	0.46%
OpenAI hosted container (Hosted Shell / Code Interpreter, 4 GB)	\$0.12 per 20-minute session → \$0.36 per hour; 1 GB \$0.03, 16 GB \$0.48, 64 GB \$1.92 per 20 minutes	\$0.36	\$0.12	4.6%	0.50%

The four accounting conventions (page 8), in one line each: output only vs all tokens — 3.2× (OSWorld-Human, \$2.43 vs \$7.87); per attempt vs per success — 4.9× (OSWorld 2.0, \$72.4 ÷ 20.6%); uncached vs cached — 10× on the read price, far less on the bill (TraceLab: 95.7% hits, prefix still 59.5% of cost); standard vs fast mode — 2× (both vendors), buying only writing speed.

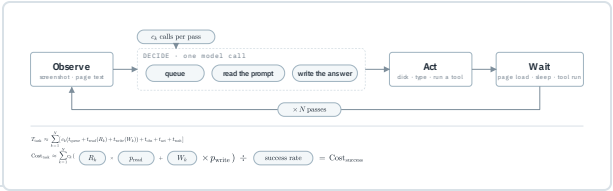
A5 · Each cause’s effect on time and on money

the complete table behind pages 9–10

CAUSE	EFFECT ON TIME	EFFECT ON MONEY	RELATIONSHIP
Many passes (N)	Roughly linear: one more step is one more call and one more wait (later steps somewhat slower, up to 3×)	Write tokens linear in N ; read tokens quadratic, because every step re-reads the whole history	same source — double the steps: about 2× the time, about 4× the read bill (uncached)
Many calls per pass (c_k)	linear	linear	same source
Reading a lot (history, screenshots, rich observations)	Reading time, longer every step	Read-token bill	same source — the same tokens cost time and money
Writing a lot (thinking)	Generation time, roughly 12 ms per token	Write-token bill at 5× the read price	same source , and the dearest time: writing is the slowest and the most expensive token
Cache hits	Saves reading time	Read price 0.1×	same source , but fragile: editing the history or switching to fast mode invalidates it
Queueing	slow	free	slow but not expensive ; escaping it means a fast or priority tier at 2×
Environment waits (page loads, fixed sleeps)	slow	No API cost; machine time by the hour — CPU sandboxes \$0.02–0.36 per hour, usually under 5% of the API bill	slow but not expensive
Tool tails (tests, builds)	slow	No API cost	slow but not expensive
Failures, idling, retries	slow	expensive	same source , and it turns “per attempt” into “per success”
Fast mode	Writing up to 2.5× faster (vendor-stated); reading unchanged	2× the price	money for time
Bigger models, more thinking	slower	dearer	money and time for accuracy , convex: marginal tokens per point 101K → 575K on one model; 6× / 9× / 9.6× tokens for a few points; more thinking not always more accurate
Smaller models	faster	cheaper	but if accuracy falls, steps and retries rise — possibly slower and dearer again (inference, no direct evidence)



A6 · The gaps in the evidence, and what each one blocks



the complete table behind page 10

GAP	WHAT EXISTS TODAY	CONSEQUENCE FOR WHAT CAN BE CLAIMED
Read / write time split and cache hit rate for a frontier API model	No measurement reports both; the only read / write split is on locally served 27–31B models (Yuan et al., 2026); screenshot-agent measurements do not report cache state	How much of “87–97% in the model” is reading, how much writing, and what remains once the cache is on — unknown
How much slower an agent is than a person	None of 17 benchmarks measures agent and human elapsed time on the same tasks; the closest is AXIS’s user study, in which a UI agent was 1.69× slower than manual work on easy tasks (small sample) (Lu et al., 2025)	“X times slower than a person” cannot go on a slide
Environment dollars	Benchmarks report instance types; AgentSysBench reports shares; products publish nothing. Computable from instance type × duration × list price: CPU environments usually 0.3–5% of the API bill (page 8)	Every benchmark cost is an API bill; the omission is small for CPU environments, not for GPU sandboxes or long idle sessions
Whether OSWorld-Human’s “action” includes the 2-s sleep after each step	Not stated (xlang-ai, 2026); if it does, “actions under 2%” implies more than 100 s per step (calc.)	“Environment under 3.5%” may understate the waiting
The rebound from a smaller model	No direct evidence on how many extra steps and retries a drop in accuracy causes	The last row of the relationship table stays marked “inference”
How much slowness hurts accuracy	Only OSWorld 2.0 records the stale-state failure mode, without its share of failures (XLANG Lab, 2026)	“It exists” can be said; “how large” cannot